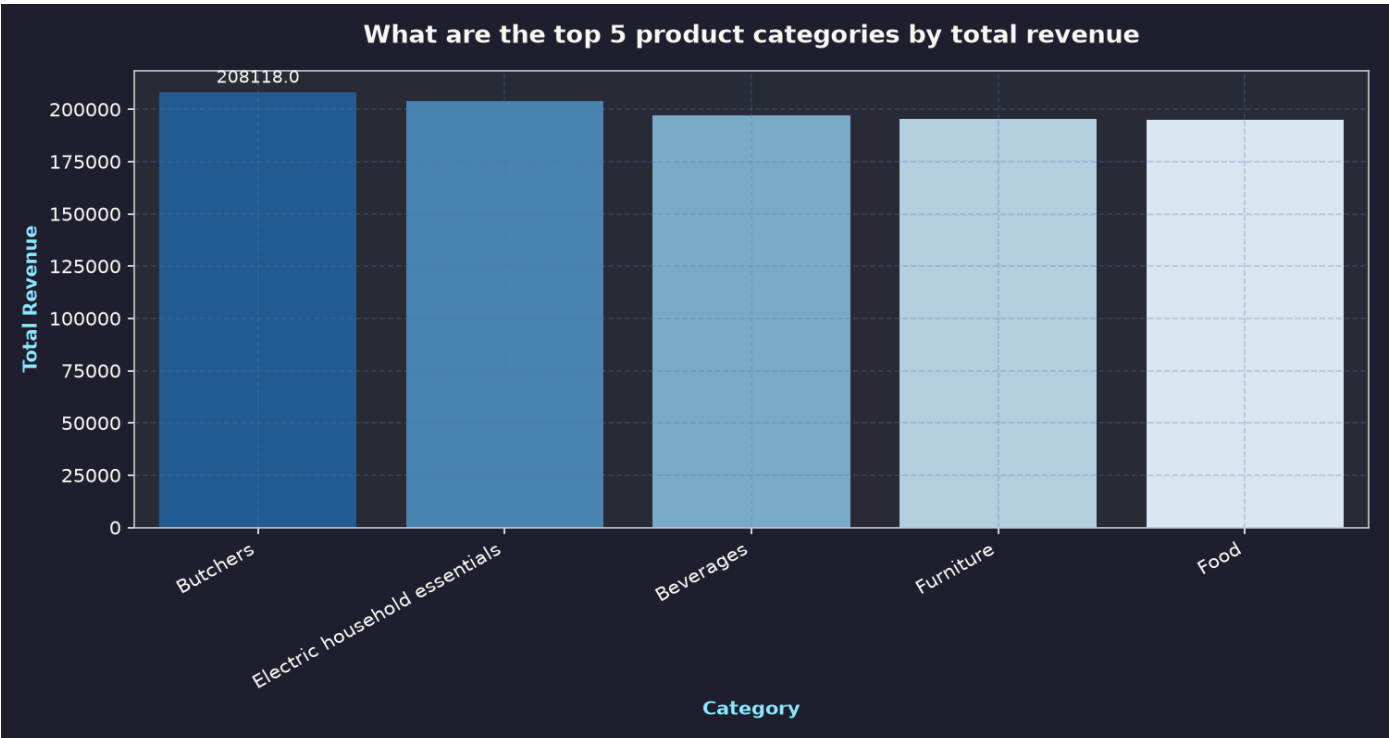


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** *What are the top 5 product categories by total revenue?*
- **Answer:** The five categories that generate the highest revenue are **Butchers**, **Electric household essentials**, **Beverages**, **Furniture**, and **Food**, with total revenues ranging from **\$194,812** to **\$208,118**.

2. Dataset Overview & Data Quality Profile

Aspect	Detail
----- -----	
Rows × Columns	12,575 × 11
Duplicate Records	0 (none found)
Missing Values	- Item - Price Per Unit - Quantity - Total Spent - Discount Applied

| **Potential Issues** | *Transaction Date* stored as a string (date-time parsing may be required for time-based analyses) |

3. Data Cleaning & Validation Summary

- **Cleaning Action:** Removed any duplicate rows (none existed) and created a clean copy of the data.
- **Validation Metrics:**
 - Clean rows after processing: **12,575** (100 % of original).
 - No duplicate rows remain.

```
# Cleaning step (already applied) result = df.drop_duplicates().copy()
```

4. User Query & Analytical Approach

- **User Query:** “What are the top 5 product categories by total revenue?”
- **Analytical Methodology:**
 1. **Group** the data by the **Category** column.
 2. **Aggregate** the **Total Spent** values within each group to compute total revenue per category.
 3. **Sort** the aggregated results in descending order of revenue.
 4. **Select** the first five rows to obtain the top-5 categories.

5. Python Analysis Code & Executed Data Output

```
# Aggregate total revenue per category and get the top 5 result = ( df.groupby('Category',
as_index=False)['Total Spent'] .sum() .rename(columns={'Total Spent': 'Total Revenue'})
.sort_values('Total Revenue', ascending=False) .head(5) )
```

Resulting Table

Category	Total Revenue
Butchers	208,118.00
Electric household essentials	203,813.50
Beverages	197,047.50
Furniture	195,310.00

| Food | 194,812.00 |

6. Key Insights & Strategic Recommendations

- **Revenue Concentration:** The top five categories alone account for a substantial share of overall sales. Prioritizing inventory, marketing, and pricing strategies for these categories can yield high ROI.
- **Butchers Lead:** With the highest revenue, the *Butchers* category may benefit from deeper margin analysis, promotional testing, and supply chain optimization.
- **Cross-Category Opportunities:** *Electric household essentials* and *Beverages* are close competitors in revenue; bundling promotions (e.g., “buy a beverage, get a discount on household essentials”) could boost average transaction value.
- **Data Quality Action:** Address the missing values in *Item*, *Price Per Unit*, *Quantity*, *Total Spent*, and *Discount Applied* to enable more granular profitability analyses (e.g., per-item margin, discount impact).
- **Date Handling:** Convert *Transaction Date* from string to proper datetime format to unlock time-series insights such as seasonal trends or month-over-month growth for these high-revenue categories.

Prepared by the Autonomous AI Data Analyst

![What are the top 5 product categories by total revenue? Chart](analysis_chart.png)